

Nominal test-1

Object 1 = A and Object 3 = C = dif = $d_i = 1$

Ordinal test-2 = fair = 1, good = 2, excellent = 3

Normalize Ord and 1 using

$$Z = \frac{P-1}{M-1} \quad \text{Where } M = 3 \text{ (No. of levels)}$$

<u>Label</u>	<u>Rank</u>	<u>Normalized</u>
fair	1	0.00
good	2	0.50
excellent	3	1.00

$$d_2 = |1.00 - 0.50| = 0.50$$

Numeric test-3 = $Z = \frac{x - \text{Min}}{\text{Max} - \text{Min}} = \text{Mean} = 22, \text{MAX} = 64$

$$\text{Object 1} = 45 \quad (45 - 22) / (64 - 22) = 23/42 = 0.5476$$

$$\text{Object 3} = 64 \quad (64 - 22) / (64 - 22) = 1.000$$

$$d_3 = |0.5476 - 1.0000| = 0.4524$$

$$D_{1,3} = \sqrt{\frac{(1)^2 + (0.5)^2 + (0.4524)^2}{3}} = \sqrt{\frac{1 + 0.25 + 0.2047}{3}}$$

$$= \sqrt{0.4847} = 0.696$$

$$\approx \underline{\underline{0.70}}$$